Marketing Funnel & Conversion Performance Analysis

Objective

Analyze customer conversion behavior from marketing campaigns and identify:

- Drop-off points
- High-performing segments
- Factors influencing conversion

Dataset

Bank Marketing Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")
```

Load Dataset

```
df=pd.read_csv(r"C:\Users\DELL\OneDrive\Desktop\DashBoard\DashBoard
PowerBi\Project 7\bank-additional-full.csv",sep=';')
df.head()
```

```
   age        job  marital    education  default housing loan
contact  \
0   56  housemaid  married     basic.4y       no      no   no
telephone
1   57   services  married  high.school  unknown      no   no
telephone
2   37   services  married  high.school       no     yes   no
telephone
3   40     admin.  married     basic.6y       no      no   no
telephone
4   56   services  married  high.school       no      no  yes
telephone

  month day_of_week  ...  campaign  pdays  previous     poutcome
emp.var.rate  \
0   may         mon  ...         1    999         0  nonexistent
1.1
1   may         mon  ...         1    999         0  nonexistent
```

```
1.1
2   may         mon  ...         1    999         0  nonexistent
1.1
3   may         mon  ...         1    999         0  nonexistent
1.1
4   may         mon  ...         1    999         0  nonexistent
1.1

   cons.price.idx  cons.conf.idx  euribor3m  nr.employed   y
0          93.994          -36.4      4.857       5191.0  no
1          93.994          -36.4      4.857       5191.0  no
2          93.994          -36.4      4.857       5191.0  no
3          93.994          -36.4      4.857       5191.0  no
4          93.994          -36.4      4.857       5191.0  no

[5 rows x 21 columns]
```

Data Understanding

```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   age             41188 non-null  int64
 1   job             41188 non-null  object
 2   marital         41188 non-null  object
 3   education       41188 non-null  object
 4   default         41188 non-null  object
 5   housing         41188 non-null  object
 6   loan            41188 non-null  object
 7   contact         41188 non-null  object
 8   month           41188 non-null  object
 9   day_of_week     41188 non-null  object
 10  duration        41188 non-null  int64
 11  campaign        41188 non-null  int64
 12  pdays           41188 non-null  int64
 13  previous        41188 non-null  int64
 14  poutcome        41188 non-null  object
 15  emp.var.rate    41188 non-null  float64
 16  cons.price.idx  41188 non-null  float64
 17  cons.conf.idx   41188 non-null  float64
 18  euribor3m       41188 non-null  float64
 19  nr.employed     41188 non-null  float64
 20  y               41188 non-null  object
```

```
dtypes: float64(5), int64(5), object(11)
memory usage: 6.6+ MB

                age      duration      campaign         pdays
previous  \
count  41188.00000  41188.000000  41188.000000  41188.000000
41188.000000
mean      40.02406    258.285010      2.567593    962.475454
0.172963
std       10.42125    259.279249      2.770014    186.910907
0.494901
min       17.00000      0.000000      1.000000      0.000000
0.000000
25%       32.00000    102.000000      1.000000    999.000000
0.000000
50%       38.00000    180.000000      2.000000    999.000000
0.000000
75%       47.00000    319.000000      3.000000    999.000000
0.000000
max       98.00000   4918.000000     56.000000    999.000000
7.000000

       emp.var.rate  cons.price.idx  cons.conf.idx     euribor3m
nr.employed
count  41188.000000    41188.000000   41188.000000  41188.000000
41188.000000
mean       0.081886       93.575664     -40.502600      3.621291
5167.035911
std        1.570960        0.578840       4.628198      1.734447
72.251528
min       -3.400000       92.201000     -50.800000      0.634000
4963.600000
25%       -1.800000       93.075000     -42.700000      1.344000
5099.100000
50%        1.100000       93.749000     -41.800000      4.857000
5191.000000
75%        1.400000       93.994000     -36.400000      4.961000
5228.100000
max        1.400000       94.767000     -26.900000      5.045000
5228.100000
```

Data Cleaning & Preparation

```python
# Convert target variable
df['conversion'] = df['y'].map({'yes': 1, 'no': 0})

# Check missing values
df.isnull().sum()
```

```
age                 0
job                 0
marital             0
education           0
default             0
housing             0
loan                0
contact             0
month               0
day_of_week         0
duration            0
campaign            0
pdays               0
previous            0
poutcome            0
emp.var.rate        0
cons.price.idx      0
cons.conf.idx       0
euribor3m           0
nr.employed         0
y                   0
conversion          0
dtype: int64
```

Key Performance Indicators (KPIs)

```
total_contacts = len(df)
total_customers = df['conversion'].sum()
conversion_rate = total_customers / total_contacts

print(f"Total Contacts: {total_contacts}")
print(f"Total Customers: {total_customers}")
print(f"Conversion Rate: {conversion_rate:.2%}")
```

```
Total Contacts: 41188
Total Customers: 4640
Conversion Rate: 11.27%
```

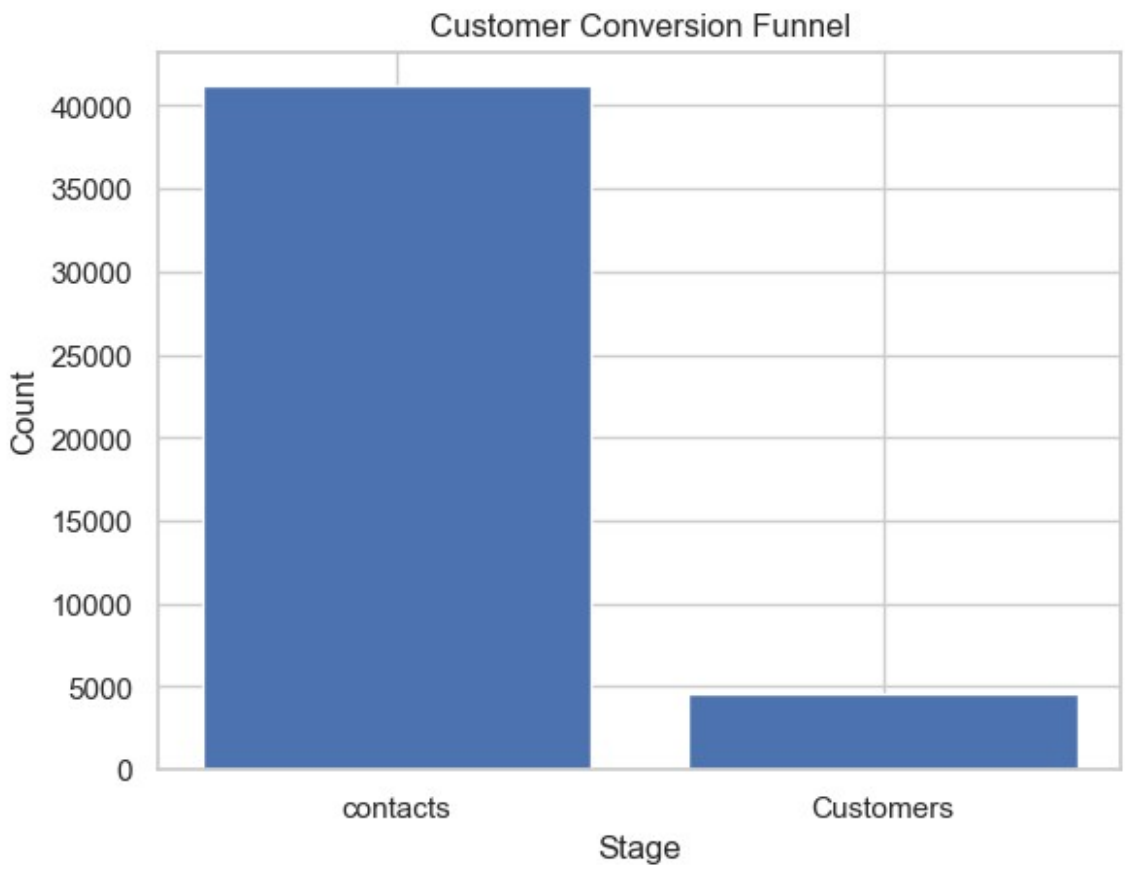
Funnel Analysis

```
funnel=pd.DataFrame({
    "Stage":["contacts","Customers"],
    "Count":[total_contacts,total_customers]})
funnel
```

```
       Stage  Count
0   contacts  41188
1  Customers   4640
```

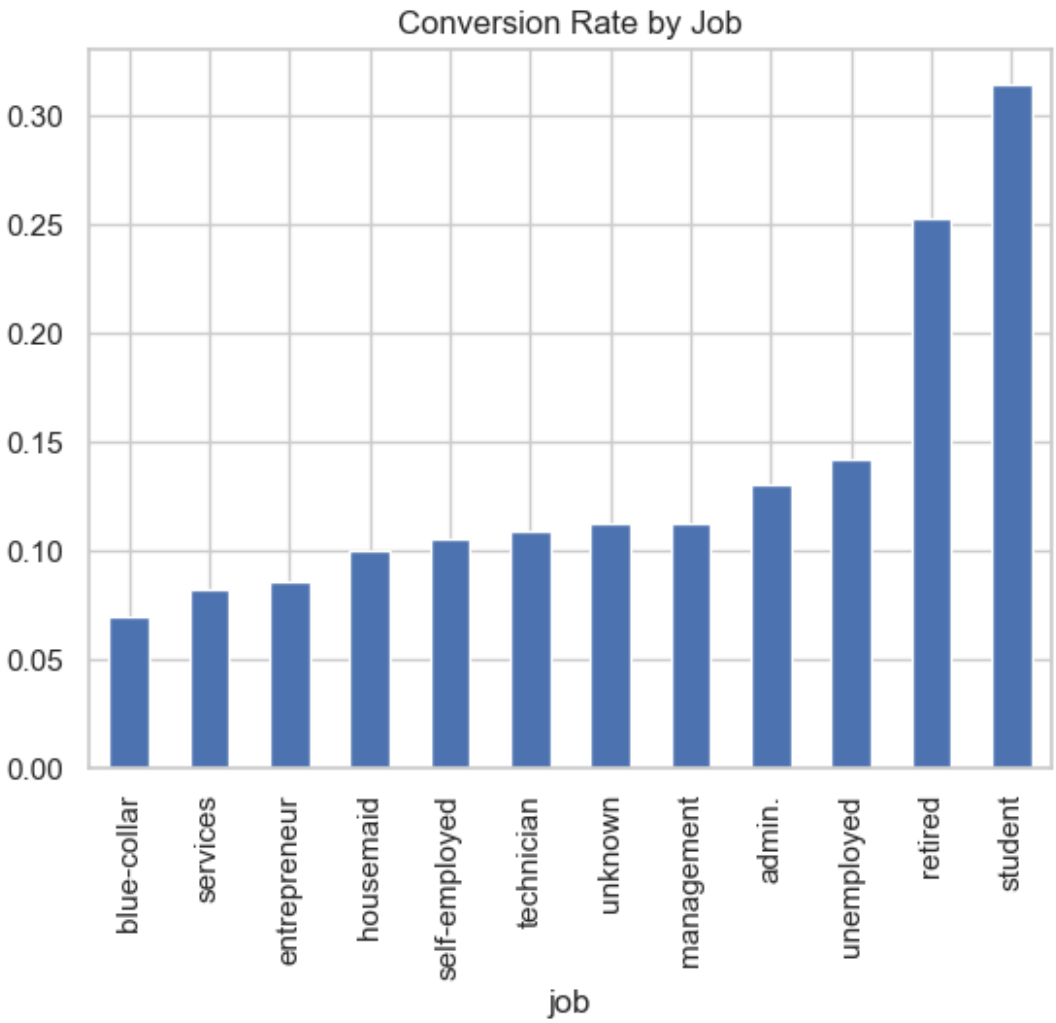
```
plt.figure()
plt.bar(funnel['Stage'],funnel['Count'])
plt.title("Customer Conversion Funnel")
plt.xlabel("Stage")
plt.ylabel("Count")
```

```
Text(0, 0.5, 'Count')
```



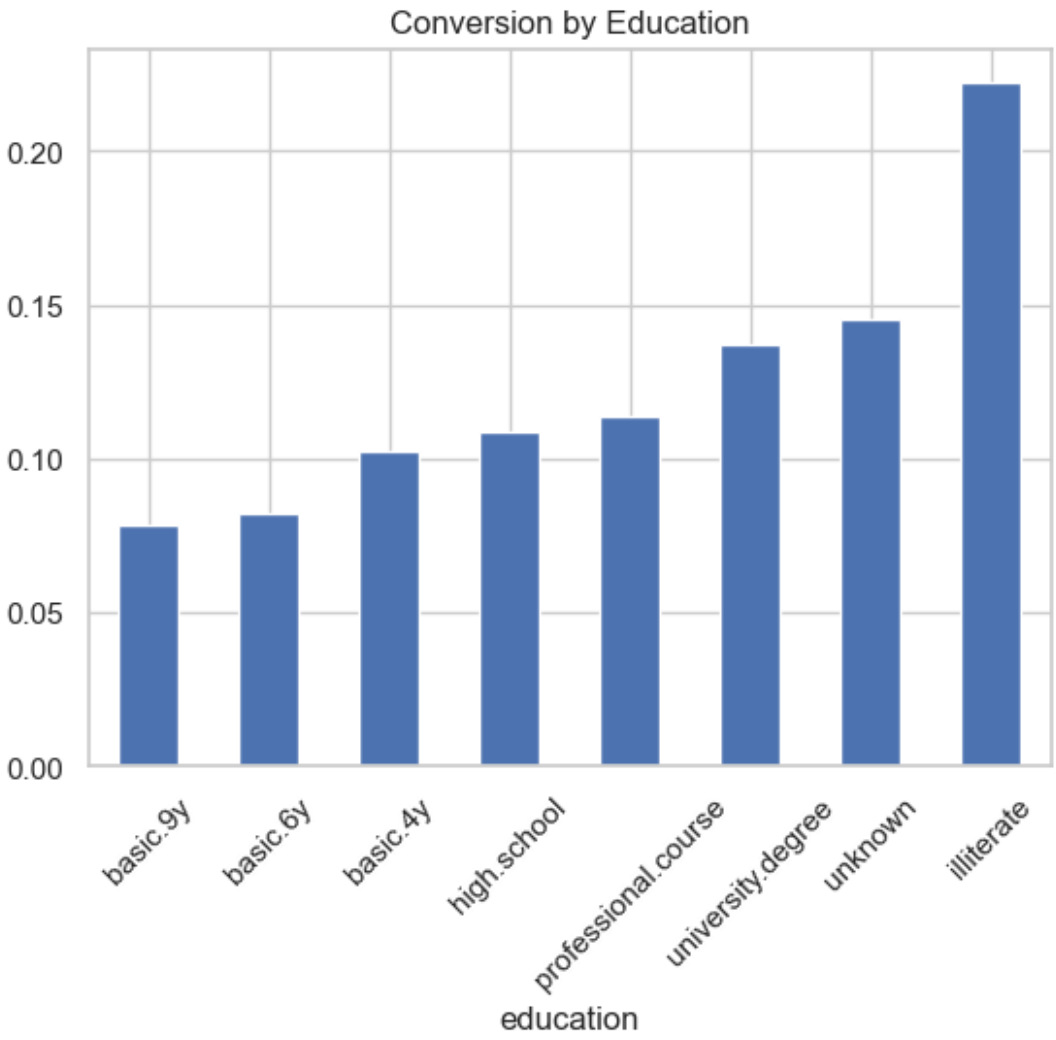
Conversion by Job Category

```
job_conv =df.groupby('job')['conversion'].mean().sort_values()
job_conv.plot(kind='bar')
plt.title("Conversion Rate by Job")
plt.show()
```



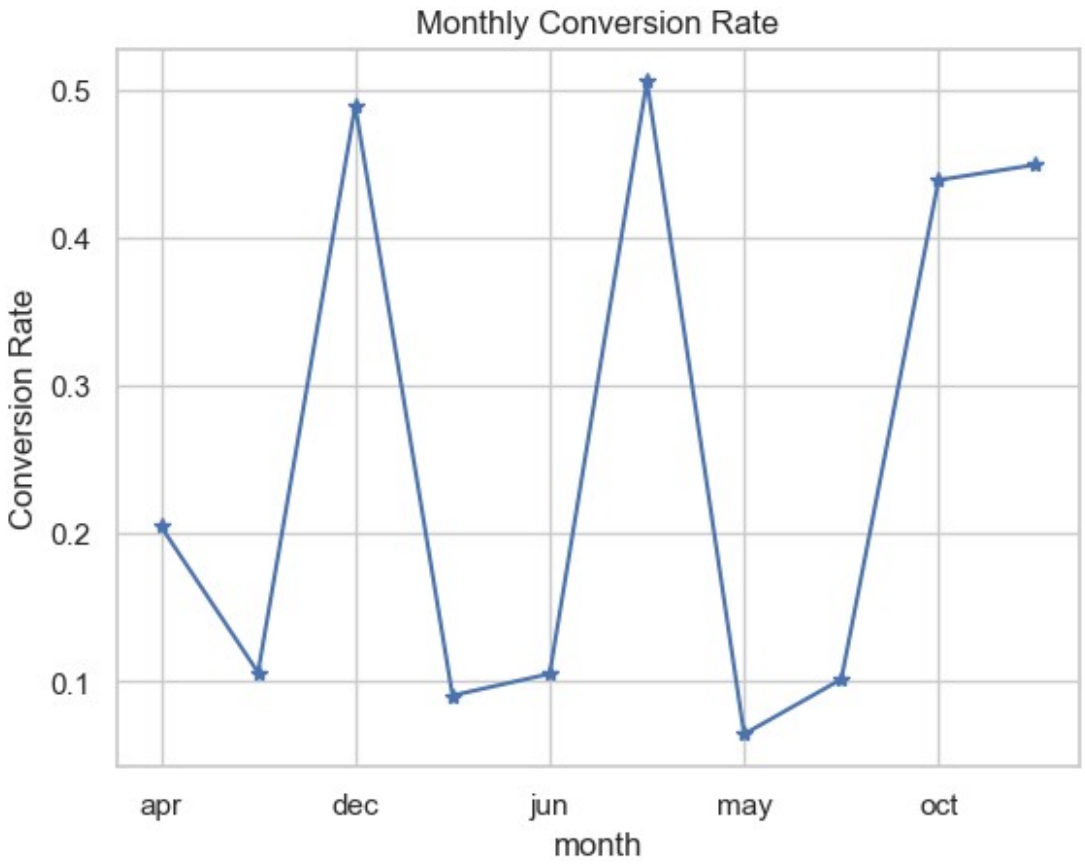
Conversion by Education Level

```
edu_conv=df.groupby('education')['conversion'].mean().sort_values()
edu_conv.plot(kind='bar')
plt.title("Conversion by Education")
plt.xticks(rotation=45)
plt.show()
```



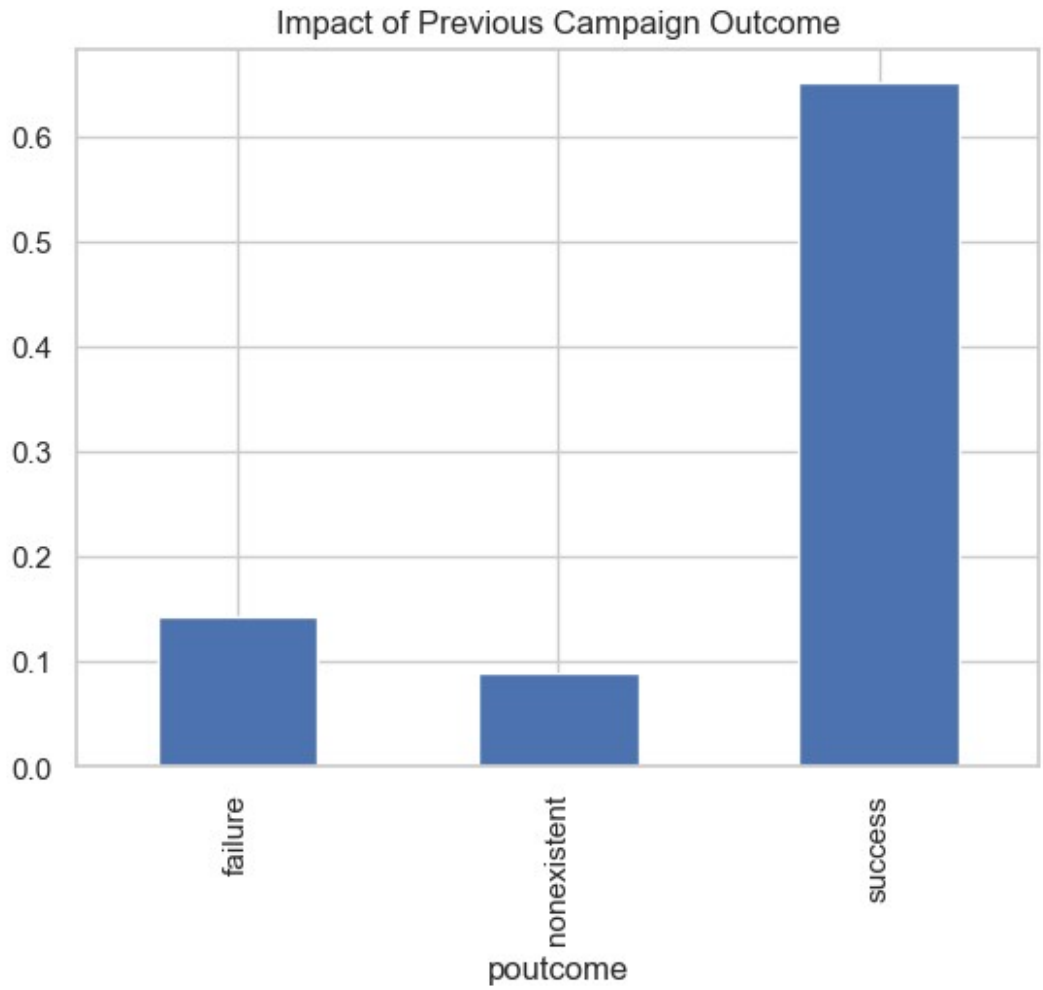
Monthly Conversion Trends

```
month_conv=df.groupby('month')['conversion'].mean()
month_conv.plot(marker='*')
plt.title("Monthly Conversion Rate")
plt.ylabel("Conversion Rate")
plt.show()
```



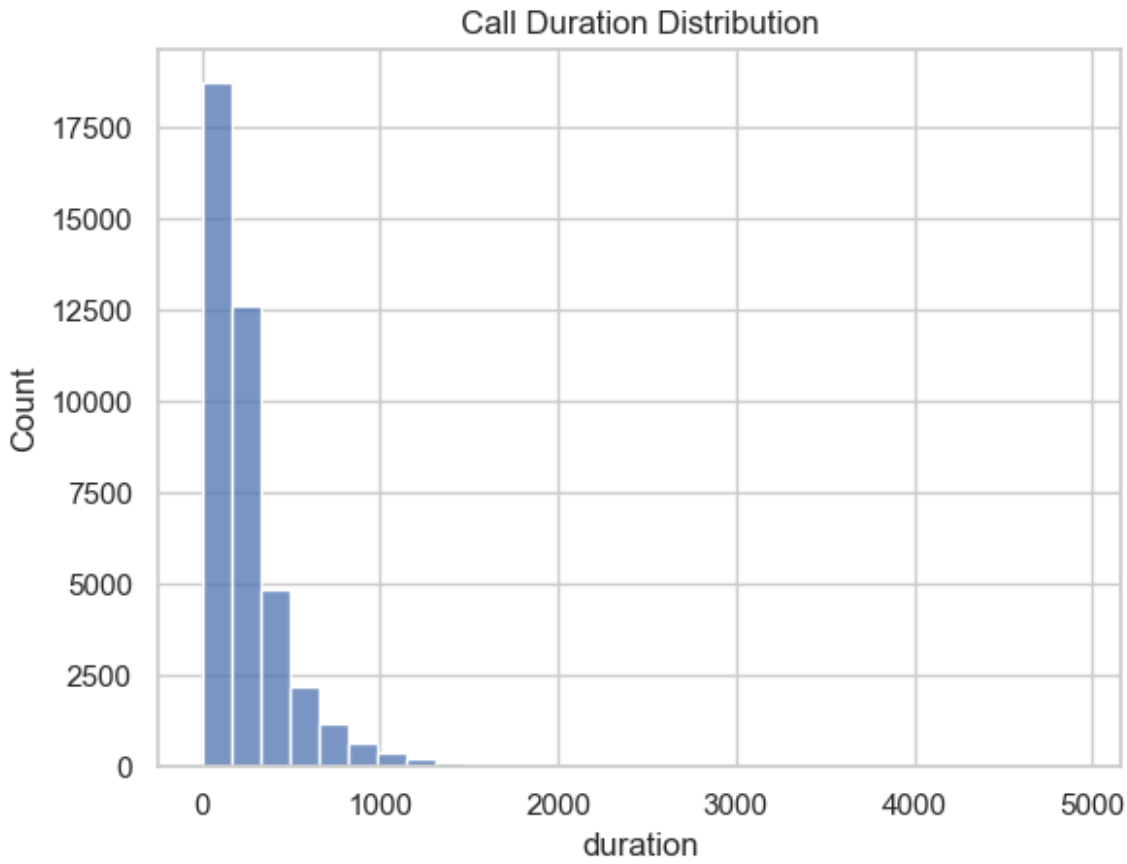
Previous Campaign Impact

```
poutcome_conv=df.groupby('poutcome')['conversion'].mean()
poutcome_conv.plot(kind='bar')
plt.title("Impact of Previous Campaign Outcome")
plt.show()
```



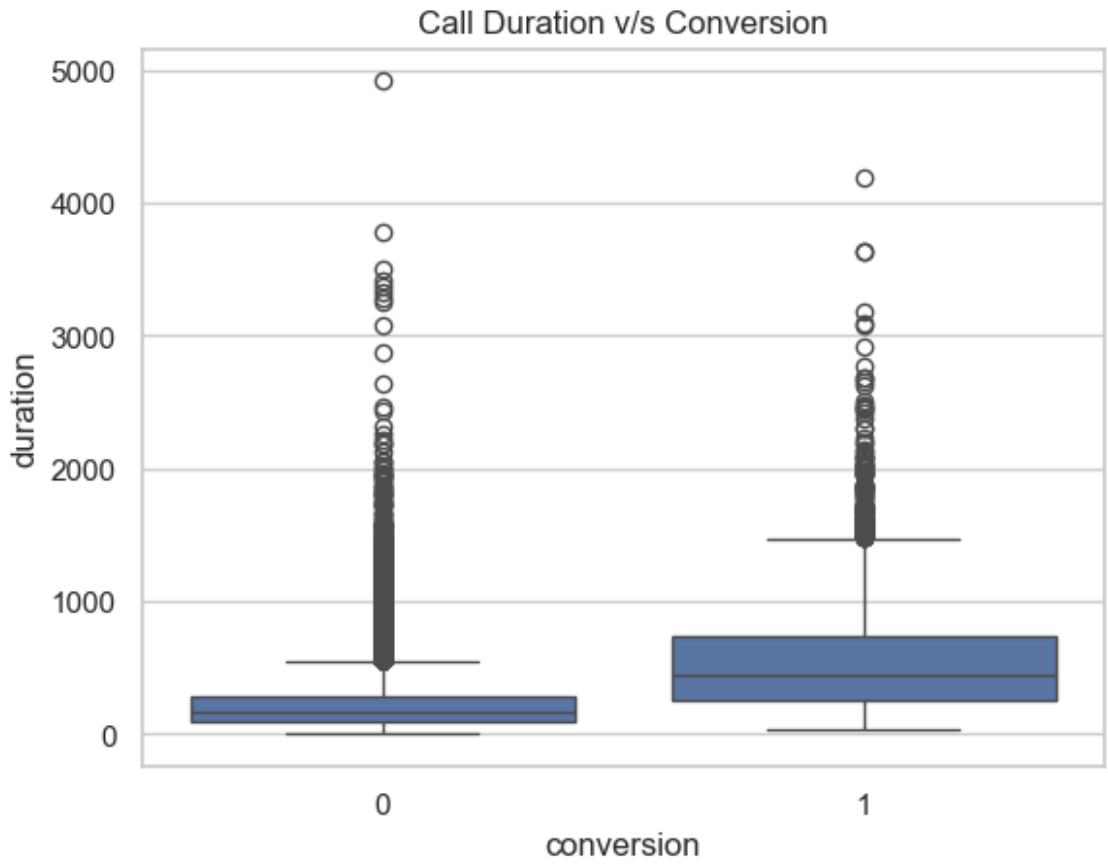
Call Duration Analysis

```
sns.histplot(df['duration'],bins=30)
plt.title("Call Duration Distribution")
plt.show()
```



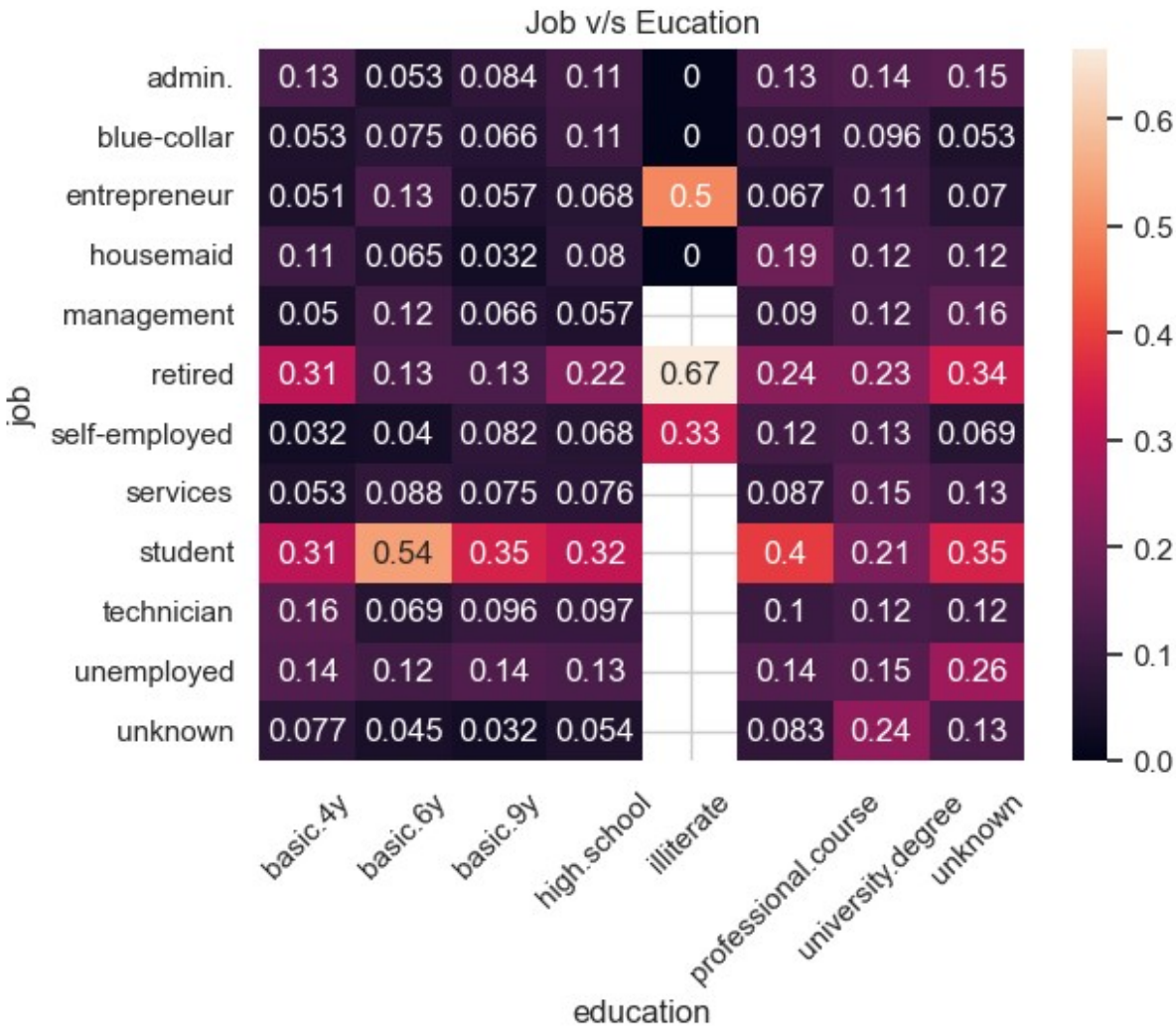
Duration v/s Conversion

```
sns.boxplot(x='conversion',y='duration',data=df)
plt.title("Call Duration v/s Conversion")
plt.show()
```



Conversion Heatmap (Job v/s Education)

```
pivot=df.pivot_table(
    values='conversion',
    index='job',
    columns='education',
    aggfunc='mean')
sns.heatmap(pivot,annot=True)
plt.title("Job v/s Eucation")
plt.xticks(rotation=45)
plt.show()
```



Advanced Behavioral Segmentation Analysis

Analyze customer behavior using segmentation and derived metrics.

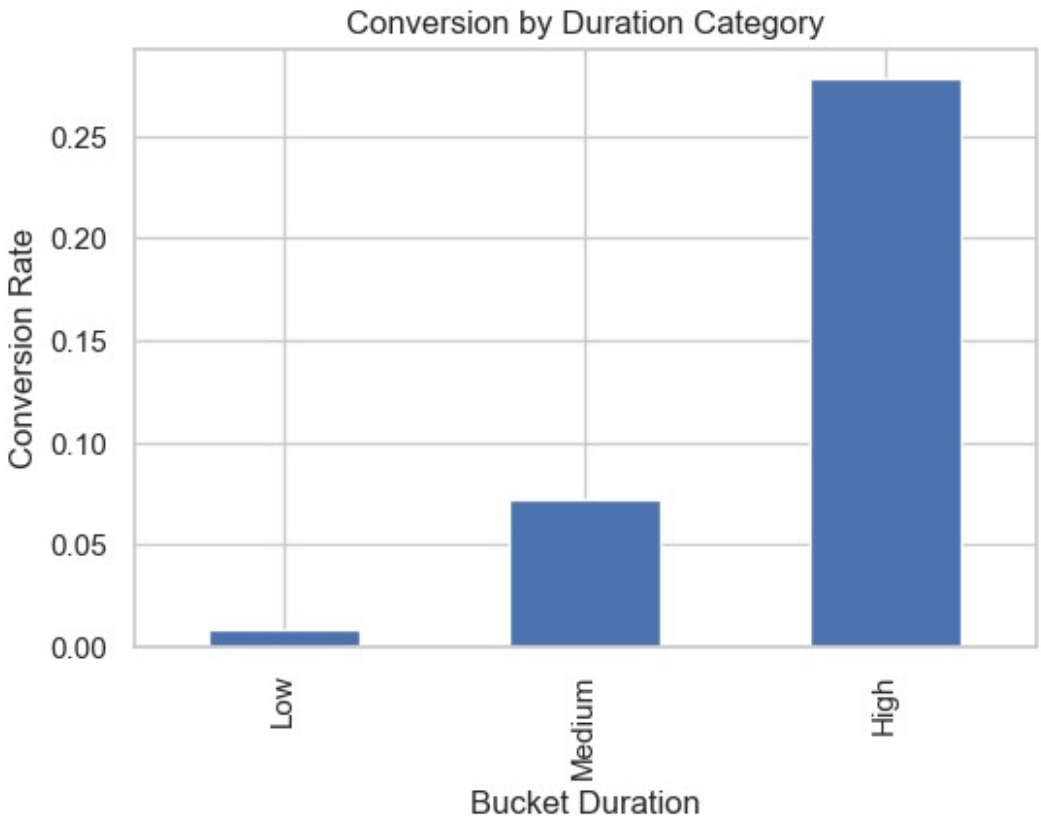
```
#creating Duration Buckets
df['duration_bucket']=pd.cut(
    df['duration'],bins=[0,100,300,10000],
    labels=['Low','Medium','High']
)
df[['duration','duration_bucket']].head(10)
```

```
   duration duration_bucket
0       261          Medium
1       149          Medium
2       226          Medium
3       151          Medium
4       307            High
5       198          Medium
```

```
6        139          Medium
7        217          Medium
8        380            High
9         50             Low
```

Conversion by Duration Category

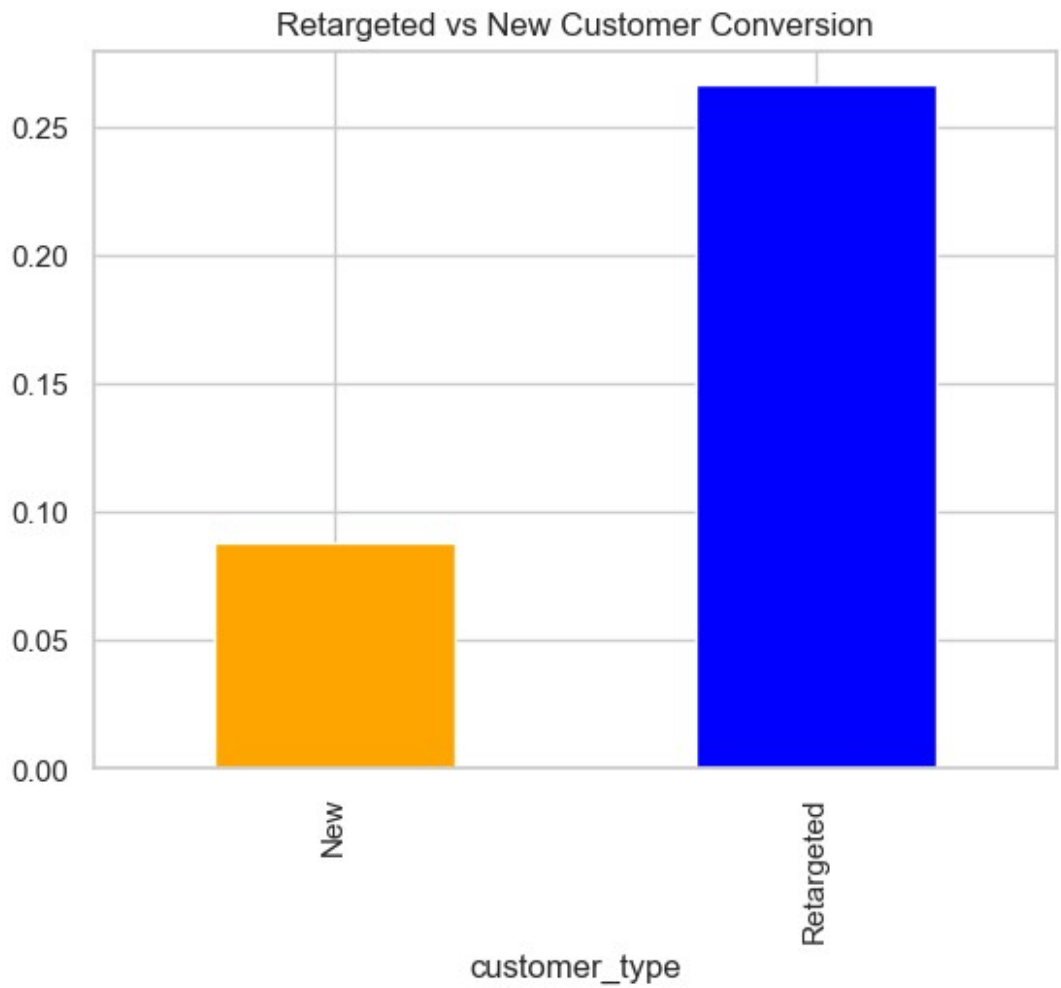
```
duration_conv=df.groupby('duration_bucket',observed=True)
['conversion'].mean()
duration_conv.plot(kind='bar',figsize=(6,4))
plt.title("Conversion by Duration Category")
plt.xlabel("Bucket Duration")
plt.ylabel("Conversion Rate")
plt.show()
```



Retargeting v/s New Customer Analysis

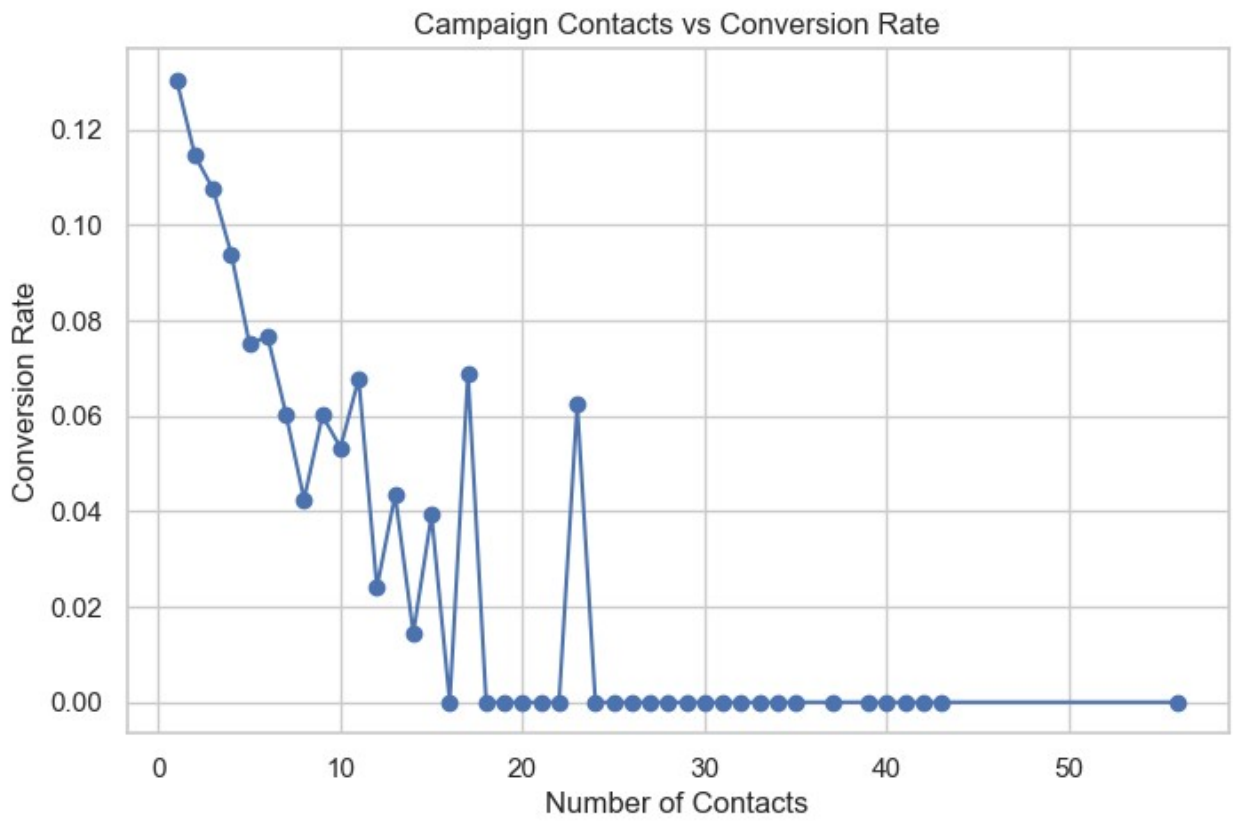
```
df['customer_type'] = np.where(df['previous'] > 0, 'Retargeted',
'New')

retarget_conv = df.groupby('customer_type')['conversion'].mean()

retarget_conv.plot(kind='bar', color=['orange','blue'])
```

```
plt.title("Retargeted vs New Customer Conversion")
plt.show()
```



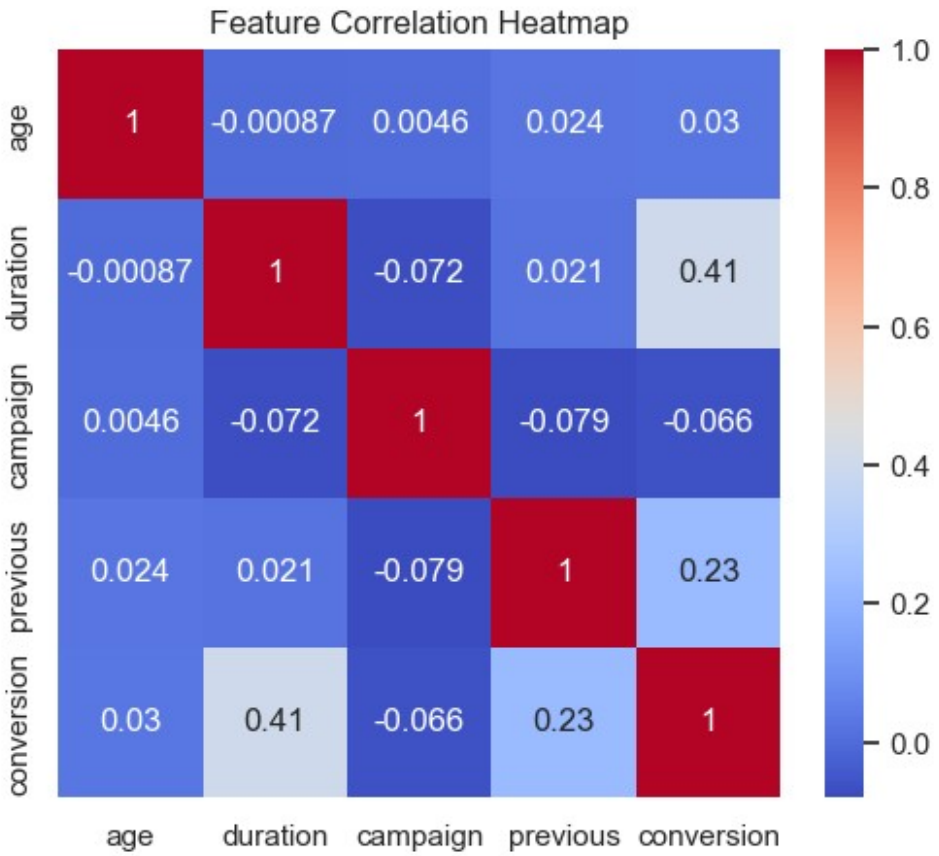
Campaign Intensity vs Conversion

```
campaign_conv = df.groupby('campaign')['conversion'].mean()

plt.figure(figsize=(8,5))
plt.plot(campaign_conv, marker='o')
plt.title("Campaign Contacts vs Conversion Rate")
plt.xlabel("Number of Contacts")
plt.ylabel("Conversion Rate")
plt.show()
```



Pairwise Feature Relationship (Correlation Heatmap)

```
corr =
df[['age','duration','campaign','previous','conversion']].corr()

plt.figure(figsize=(6,5))
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Feature Correlation Heatmap")
plt.show()
```



Advanced Insights

Advanced Insights

Conversion Distribution Across Age Groups

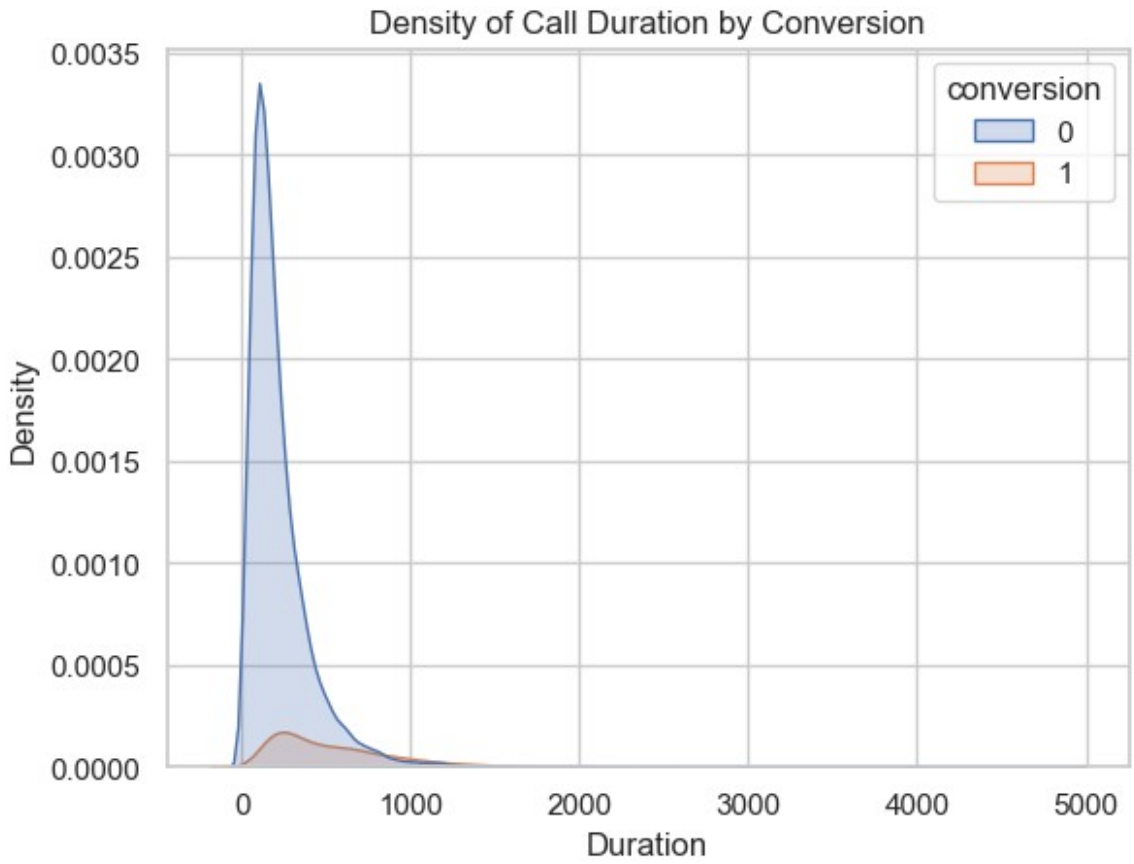
df['age_group'] = pd.cut(df['age'], bins=[18,30,45,60,100], labels=['18-30','30-45','45-60','60+'])

age_conv = df.groupby('age_group',observed=True)['conversion'].mean()

age_conv.plot(kind='bar') plt.title("Conversion Rate by Age Group") plt.show()

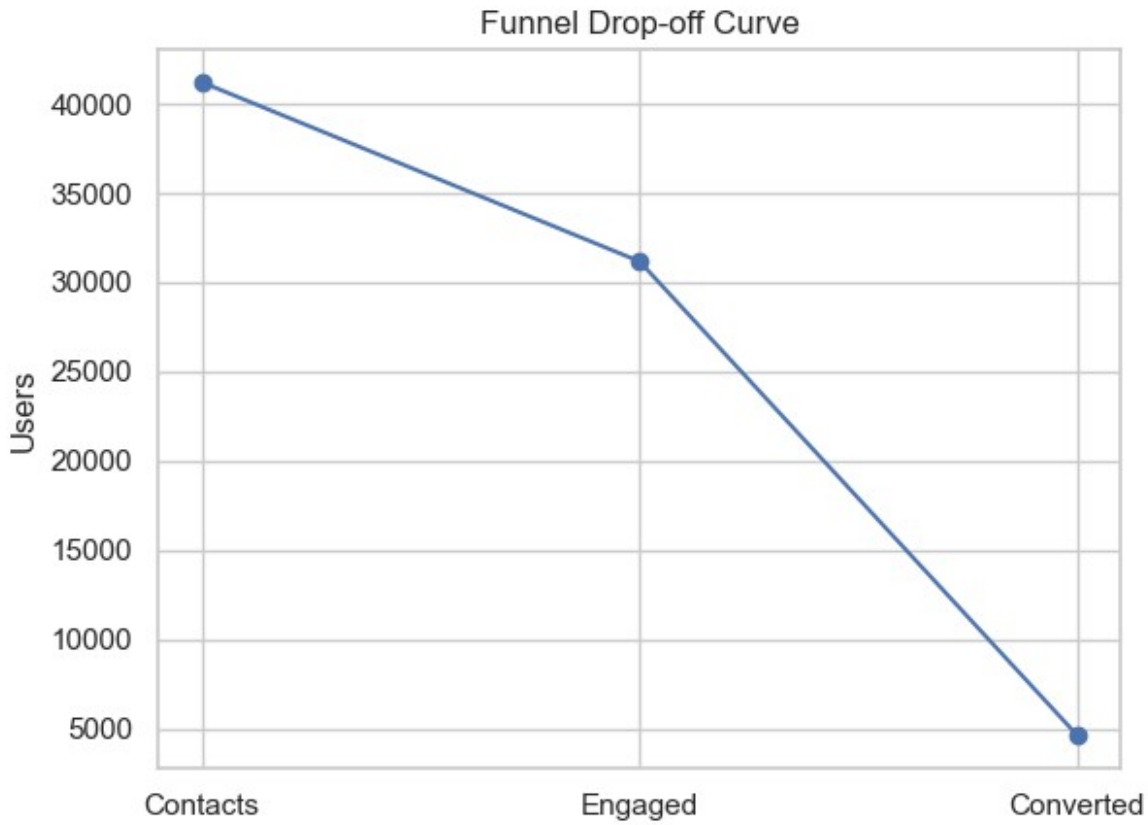
Probability Density of Call Duration (Converted vs Not)

```
sns.kdeplot(data=df, x='duration', hue='conversion', fill=True)
plt.title("Density of Call Duration by Conversion")
plt.xlabel('Duration')
plt.show()
```



Drop-off Simulation Curve

```
drop_stages = ['Contacts', 'Engaged', 'Converted']
values = [len(df), len(df[df['duration']>100]),
df['conversion'].sum()]

plt.plot(drop_stages, values, marker='o')
plt.title("Funnel Drop-off Curve")
plt.ylabel("Users")
plt.show()
```



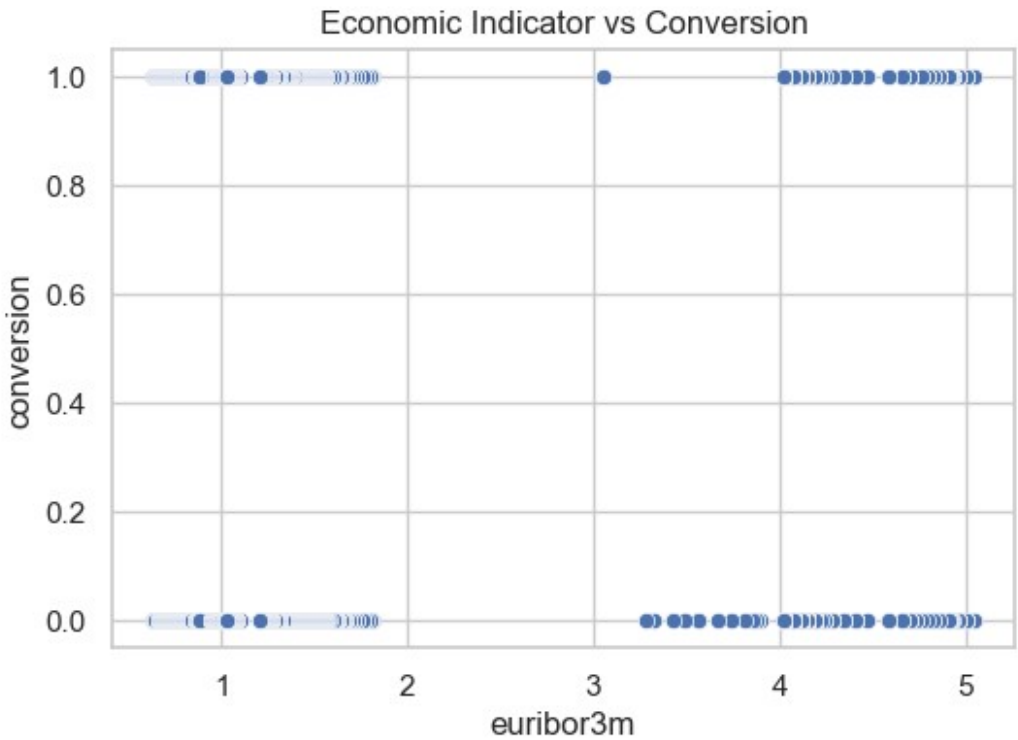
Customer Risk Segmentation

Identify low probability customers.

```
risk_customers = df[(df['duration'] < 100) & (df['campaign'] > 3)]
print("High Risk Customers:", len(risk_customers))

High Risk Customers: 2767
```

Conversion by Economic Indicators

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='euribor3m', y='conversion', data=df)
plt.title("Economic Indicator vs Conversion")
plt.show()
```



Top Performing Segments

```
top_segments = df.groupby(['job','education'])
['conversion'].mean().sort_values(ascending=False).head(10)
top_segments
```

```
job            education
retired        illiterate             0.666667
student        basic.6y               0.538462
entrepreneur   illiterate             0.500000
student        professional.course    0.395349
               basic.9y               0.353535
               unknown                0.353293
retired        unknown                0.336735
self-employed  illiterate             0.333333
student        high.school            0.319328
retired        basic.4y               0.309883
Name: conversion, dtype: float64
```